SUPPLY INLETS

J4
FAN1_12V
FAN1 12V IN
J6
FAN2_12V
FAN2 12V IN
J3
PWR IN Galvo
12.6V IN (BUCK)
VIN_RAW
Pin_1
Pin_2
D2
1N5822
VIN_BUCK

BUCK CONVERTER LMR33630 – 12.6V INPUT TO +5V

U_BUCK1
LMR33630ADDA
VIN_BUCK
EN_BUCK
VCC_BUCK
BOOT_BUCK
SW_BUCK
FB_BUCK
C_IN1
10uF
C_IN2
10uF
C_INHF1
220nF
C_BOOT1
100nF
C_VCC1
1uF
R_EN1
63.4k 1%
R_EN2
10k 1%
R_FB1
100k 1%
R_FB2
5.9k 1%
D1
1N5819
L_BUCK1
6.8uH
C_OUT1
22uF
C_OUT2
22uF

+5V ANALOG SUPPLY FILTER

+5V_MCU
FB_ANA
SW_ANA
MP18PG2215N1

RAIL FLAGS (ERC)

PWR_FLAG
Buck +5V
5V GND Buck
AGND (from Galvo Board)
+3V3
+15V
-15V
VIN_RAW
VIN_BUCK
+5V_ANA
+5V_MCU

SD CARD (SPI3)

J1
SD Card Module (6-pin 2.54mm)
SD_CS
SD_MOSI
SD_SCK
SD_MISO

TEMPERATURE SENSORS – DS18B20 1-WIRE

U6
U7
U8
U9
U10
DS18B20 connector
R5
4.7K

ESP32-S3-N16R8 CONTROLLER

U1
ESP32-S3-N16R8-DevKit
GPIO4
GPIO7
GPIO8
GPIO14
GPIO21
GPIO38
GPIO39
GPIO47
SD_SCK
SD_MOSI
SD_MISO
SD_CS
FAN1_PWM
FAN2_PWM
FAN1_TACH
FAN2_TACH
DAC_SCLK_3V3
DAC_DIN_3V3
DAC_SYNC_3V3
DAC_CLR_3V3
+5V_MCU

FAN CONTROL + TACHO

J5
FAN1
J7
FAN2
R30
4.7K
R31
4.7K

DMX

J_DMX1
DMX MAX485
DMX_RO_5V
R_DMXIN1
18k
C_DMXLV1
100nF
U_DMXLV1
SN74LVC1G17DBVR
3V3 DMX input protection: no galvanic isolation

DAC INPUT DEFAULTS / 3V3

R_DACCLRA1
DAC_CLR_3V3
R_DACDINA1
DAC_DIN_3V3
R_DACSCLKA1
DAC_SCLK_3V3
R_DACSYNCA1
DAC_SYNC_3V3
10k
U_5V_ANA
SN74LVC8T245PWR
C_DACLVB1
100nF

DAC8562 – DUAL 16-BIT DAC (SPI2)

U2
DAC8562SDGST
C_ANA1
C_ANA2
10uF
100nF
R_DACSCLKB1
R_DACDINB1
R_DACSYNCB1
R_DACCLRB1
R_DACUNUSED1
C1
C2
C3
C4
100nF
1nF
R13
100
X_NODE1
V_NODE1
U12
OPA4134UA/2K5

GALVO ANALOG OUTPUT – OPA4134

R9
R10
R11
R12
R14
R15
R16
R17
R25
R26
22k
10k
100
VREF_BUF
U13
Galvo Input

DAC OUTPUT DEFAULTS / 5V

LASER TTL INTERFACE – COMMON GROUND (INVERTED)

R27
R28
R29
10k
GPIO7
GPIO8
GPIO21
R1
R2
R4
220
U15
U16
U17
SN7407
TTL_RGB_RED
TTL_RGB_GREEN
TTL_RGB_BLUE
C5
C6
C7
100nF
R18
R20
R22
1k
R19
R21
R23
U3
TTL RGB Blue

RGB PULL-UPS – default-OFF not yet verified

EXTERNAL INTERFACES – E-STOP / DMX / SSR

J_ESTOP1
E_STOP
GPIO47
J_SSR1
SSR1 CTRL
R_SSRCTRL1
330
3V3 scan-status input protection; not a safety interlock

HARDWARE WATCHDOG – NE555

C_TRIGC_WD1
D_TRIGCL_WD1
R_TRIGPU_WD1
WD_TRIG
WD_CONT
WD_RC
WD_OUT
C_WD1
C_WDVCC1
R_WDCONT1
R_WDRC1
U_WD1
NE555D
1N4148W

SCAN-FAIL DETECTOR – NE555

SCAN_TRIG
SCAN_CONT
SCAN_RC
SCAN_STATUS_5V
C_SCANX1
C_SCANY1
D_TRIGCL_SCAN1
R_TRIGPU_SCAN1
R_SCANIN1
R_SCANRC1
C_SCANVCC1
C_SCANLV1
C_SCANCONT1
U_SCANLV1
U_SCAN1
SN74LVC1G17DBVR

NOT FOR LASER OPERATION
ENGINEERING DRAFT – NOT FOR FABRICATION

Sheet: /
File: Laser Controllerv2_only_for_pcb_test.kicad_sch

Title: GalvOS ESP32 Laser Controller – V2 engineering review

Size: User	Date: 2026-09-14	Rev: V2.0.10 DRAFT
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